

Printer API

Development Documentation

V1.1.0

Update Records

Date	Description
2022-10-31	First Edition Release
2023-02-22	Modify the interface
2023-04-23	Update the interface
2024-06-20	Update API and documentation

Contents

- 1 Introduction
 - 1.1 Detect the printer type
 - 1.2 UsbThermalPrinter Class
 - 1.2.1 UsbThermalPrinter
 - 1.2.2 Get the Model Version
 - 1.2.3 Get the printer status
 - 1.2.4 Printer reset
 - 1.2.5 Look for black labels
 - 1.2.6 Paper Feed
 - 1.2.7 Set the print alignment
 - 1.2.8 Set the Left Margin
 - 1.2.9 Set the Line Spacing
 - 1.2.10 Font Bold
 - 1.2.11 Set the Font Size
 - 1.2.12 Set the Print Grayscale Value
 - 1.2.13 Add Print Content
 - 1.2.14 Start Printing
 - 1.2.15 Get the Total Width of the Input Text
 - 1.2.16 Adds an inline offset print
 - 1.2.17 Add contents of the current printed line at the end
 - 1.2.18 Print picture/Bar code/QR code
 - 1.2.19 Table printing, printString is not executed
 - 1.2.20 Set italics
 - 1.2.21 Set the print paper type
 - 1.2.22 Adapt the new label paper
 - 1.2.23 Feed out the label paper
 - 1.2.24 Roll back the label paper
 - 1.3 ThermalPrinter Class
 - 1.3.1 Printer width
 - 1.3.2 Printer language type
 - 1.3.3 Printer status
 - 1.3.4 Initialize the 80mm USB printer
 - 1.3.5 Initialize the 80mm serial printer
 - 1.3.6 Printer Resource Request
 - 1.3.7 Get the Model Version
 - 1.3.8 Printer Reset
 - 1.3.9 Look for Black Labels
 - 1.3.10 Paper Feed
 - 1.3.11 Set the Print Alignment
 - 1.3.12 Set the Font Size
 - 1.3.13 The Print Font Width and Height are Enlarged
 - 1.3.14 Set the Print Grayscale Value

- [1.3.15 Add Print Content](#)
- [1.3.16 Start Printing](#)
- [1.3.17 Paper Cut](#)
- [1.3.18 Send Instructions to the Printer Device](#)
- [1.3.19 Print the Picture](#)
- [1.3.20 Release the Printer](#)
- [1.3.21 Set the Anti-white](#)
- [1.3.22 Set the paper width](#)
- [1.3.23 Get the print count](#)
- [1.3.24 Get printer status](#)

1 Introduction

1.1 Detect the printer type

```
int printerCheck = SystemUtil.checkPrinter581(this);
if(printerCheck == PRINTER_80MM_USB_COMMON){
    //SY581 USB printer
}else if(printerCheck == PRINTER_SY581){
    //SY581 Serial printer
}else if(printerCheck == PRINTER_SY581){
    //58mm Serial/USB printer
}
```

1.2 UsbThermalPrinter Class

Package: com.common.apiutil.printer

Thermal USB/Serial printer interface(56 mm)

1.2.1 UsbThermalPrinter

```
/**
 * @Function UsbThermalPrinter
 *
 * @Parameter context
 *
 * */

public UsbThermalPrinter(Context context)
```

1.2.2 Get the Model Version

```
/**
 * @Function Get the model version
 *
 * @Return Model version
 *
 * @Exception CommonException
 *
 * */

public synchronized String getVersion() throws CommonException
```

1.2.3 Get the printer status

```

/**
 * @Function Get the printer status
 *
 * @Return ResultCode: please refer the quick guidance document.
 *
 * @Exception CommonException
 *
 * */

public synchronized static int checkStatus() throws CommonException

```

1.2.4 Printer reset

```

/**
 * @Function The printer resets, all settings return to their default settings,
and the print buffer is cleared
 *
 * @Exception CommonException
 *
 * */

public synchronized void reset() throws CommonException

```

1.2.5 Look for black labels

```

/**
 * @Function Look for black labels
 *
 * @Parameter
 * 1.search_disdance: Find the maximum number of paper lines before the black
label, 0-255
 * 2.walk_disdance: The number of paper lines after the black label is found, 0-
255
 *
 * @Exception CommonException
 *
 * */

public synchronized void searchMark(int search_disdance, int walk_disdance) throws
CommonException

```

1.2.6 Paper Feed

```

/**
 * @Function Paper Feed
 *

```

```

* @Parameter
* 1.line: int > 0
*
* @Exception CommonException
*
* */

public synchronized void walkPaper(int line) throws CommonException

```

1.2.7 Set the print alignment

```

/**
 * @Function Set the print alignment
 *
 * @Parameter
 * 1.mode:
 *     1.ALGIN_LEFT
 *     2.ALGIN_MIDDLE
 *     3.ALGIN_RIGHT
 *
 * @Exception CommonException
 *
 * */

public synchronized void setAlgin(int mode) throws CommonException

```

1.2.8 Set the Left Margin

```

/**
 * @Function Set the left margin
 *
 * @Parameter
 * 1.leftDistance: Margin value: 0-255, Unit: 0.125mm
 *
 * @Exception CommonException
 *
 * */

public synchronized void setLeftIndent(int leftDistance) throws CommonException

```

1.2.9 Set the Line Spacing

```

/**
 * @Function Set the line spacing
 *
 * @Parameter

```

```
* 1.lineDistance: 0-255
*
* @Exception CommonException
*
* */
```

```
public synchronized void setLineSpace(int lineDistance) throws CommonException
```

1.2.10 Font Bold

```
/**
 * @Function Font bold
 *
 * @Parameter
 * 1.isBold: True: font bold; False: font not bold
 *
 * @Exception CommonException
 *
 * */
```

```
public synchronized void setBold(boolean isBold) throws CommonException
```

1.2.11 Set the Font Size

```
/**
 * @Function Set the font size
 *
 * @Parameter
 * 1.size: 8-64
 *
 * @Exception CommonException
 *
 * */
```

```
public synchronized void setTextSize(int size) throws CommonException
```

1.2.12 Set the Print Grayscale Value

```
/**
 * @Function Set the print grayscale value
 *
 * @Parameter
 * 1.gray: Need to check if the hardware supports it
 *
 * @Exception CommonException
 *
```



```
* */  
  
public synchronized void setGray(int gray) throws CommonException
```

1.2.13 Add Print Content

```
/**  
 * @Function Add print content  
 *  
 * @Parameter  
 * 1.content: print content  
 *  
 * @Exception CommonException  
 *  
 * */  
  
public synchronized void addString(String content) throws CommonException
```

1.2.14 Start Printing

```
/**  
 * @Function Start printing, and the paper will not be taken after printing  
 *  
 * @Exception CommonException  
 *  
 * */  
  
public synchronized void printString() throws CommonException
```

1.2.15 Get the Total Width of the Input Text

```
/**  
 * @Function Gets the total width of the input tex  
 *  
 * @Parameter  
 * 1.text: text content  
 *  
 * @Exception CommonException  
 *  
 * */  
  
public synchronized int measureText(String text) throws CommonException
```

1.2.16 Adds an inline offset print

```

/**
 * @Function Adds an inline offset print
 *
 * @Parameter
 * 1.offset: Inline offset, which must be an integer multiple of 8, cannot be less
than the left margin, and cannot be greater than (384-right margin)
 * 2.content: String content
 *
 * @Exception CommonException
 *
 * */

public synchronized void addStringOffset(int offset, String content) throws
CommonException

```

Tips: This function is suitable for USB printing two stages, USB printing one stage is not valid

- 1. Note that this method can only be added with content within one line, and characters beyond one line will be intercepted and not printed
- 2. When adding content within the same line, offset needs to be added from small to large
- 3. When content is added within the same line, the print format is not allowed in the middle, otherwise it will cause word wrapping
- 4. During the same line addition process, calling the "addString()" method automatically wraps
- 5. Add the contents of this line at the end of the line and call the "endLine()" method

1.2.17 Add contents of the current printed line at the end

```

/**
 * @Function Adds the contents of the current printed line to wrap after the
addStringOffset method call
 *
 * @Exception CommonException
 *
 * */

public synchronized void endLine() throws CommonException

```

1.2.18 Print picture/Bar code/QR code

```

/**
 * @Function Print picture/Bar code/QR code
 *
 * @Parameter
 * 1.image: Bitmap objects, the width must be less than 384 pixels and an integer
multiple of 8, and the high pixels must be an integer of 8
 * 2.isBuffer: Whether to join the buffer queue, call printString or
printStringAndWalk, the buffer queue is used. There is data printed together,

```

```

mostly used for mixed printing of graphics and text
*
* @Exception CommonException
*
* */

public synchronized void printLogo(Bitmap image, boolean isBuffer) throws
CommonException

//Bar code example
Bitmap bitmap = CreateCode(barcodeStr, BarcodeFormat.CODE_128, 320, 176);
if (bitmap != null) {
    mUsbThermalPrinter.printLogo(bitmap, true);
}

```

1.2.19 Table printing, printString is not executed

```

/**
 * @Function Table printing, printString is not executed
 *
 * @Parameter
 * 1.colsTestArr: An array of column text strings
 * 2.colsWidthArr: The width weight of each column is the proportion of each
column
 * 3.colsAlign: Column alignment: 0 Left, 1 center, 2 right
 * 4.colsTextSize: Font size 12/24/36
 *
 * @Exception CommonException
 *
 * */

public synchronized void addColumnsString(String[] colsTestArr, int[]
colsWidthArr, int[] colsAlign, int colsTextSize) throws CommonException

```

1.2.20 Set italics

```

/**
 * @Function Set italics
 *
 * @Parameter isItalic
 *
 * @Exception CommonException
 *
 * */

public void setItalic(boolean isItalic) throws CommonException

```

1.2.21 Set the print paper type

```

/**
 * @Function Set the print paper type
 *
 * @Parameter
 * 1.mode: 0, normal paper; 2, label paper; default: label paper
 *
 * @Exception CommonException
 *
 * */

public synchronized void setAlgorithm(int mode) throws CommonException

```

1.2.22 Adapt the new label paper

```

/**
 * @Function Adapt the new label paper
 *
 * @Parameter
 * 1.line: line number(mm)
 *
 * @Exception CommonException
 *
 * */

public synchronized void setLableAdapt(int line) throws CommonException

```

1.2.23 Feed out the label paper

```

/**
 * @Function Feed out the label paper
 *
 * @Parameter
 * 1.line: line number(mm)
 *
 * @Exception CommonException
 *
 * */

public synchronized void setLableFeedOutNextGap(int line) throws CommonException

```

1.2.24 Roll back the label paper

```

/**
 * @Function Roll back the label paper
 *

```

```
* @Parameter
* 1.line: line number(mm)
*
* @Exception CommonException
*
* */
```

```
public synchronized void rollback(int line) throws CommonException
```

1.3 ThermalPrinter Class

Package: com.common.apiutil.printer

Thermal USB/Serial printer interface(80mm)

1.3.1 Printer width

Type	Description
ThermalPrinter.PAPER_80mm	80mm
ThermalPrinter.PAPER_58mm	58mm

1.3.2 Printer language type

Type	Description
ThermalPrinter.ENGLISH	English
ThermalPrinter.PERSIAN	Persian
ThermalPrinter.KOREAN	Korean
ThermalPrinter.JAPANESE	Japanese
ThermalPrinter.FRENCH	French
ThermalPrinter.RUSSIAN	Russian
ThermalPrinter.THAILAND	Thailand
ThermalPrinter.CAMBODIA	Cambodia
ThermalPrinter.LAOS	Laos
ThermalPrinter.CHINESE	Chinese
ThermalPrinter.PORTUGUESE	Portuguese
ThermalPrinter.SPANISH	Spanish
ThermalPrinter.ITALIAN	Italian
ThermalPrinter.GERMAN	German
ThermalPrinter.HONGKONG_CHINA	China HongKong

1.3.3 Printer status

Type	Description
ThermalPrinter.STATUS_OK	The status is normal
ThermalPrinter.STATUS_NO_PAPER	No paper
ThermalPrinter.STATUS_OVER_HEAT	Over heat
ThermalPrinter.STATUS_OVER_FLOW	The cache is full
ThermalPrinter.STATUS_UNKNOWN	Unknown error
ThermalPrinter.STATUS_BOX_OPEN	The printer box is opened
ThermalPrinter.STATUS_CUT_WRONG	The printer cutter is incorrect

1.3.4 Initialize the 80mm USB printer

```
/**
 * @Function Initialize the 80mm USB printer
 *
 * */

public synchronized static boolean init80mmUsbPrinter(Context context)
```

1.3.5 Initialize the 80mm serial printer

```
/**
 * @Function Initialize the 80mm serial printer
 *
 * */

public synchronized static boolean init80mmSerialPrinter()
```

1.3.6 Printer Resource Request

```
/**
 * @Function Printer resource request
 *
 * @Tips Printer resource application, you must first apply to use the printer-
related operations, after using the stop() method to release
 *
 * @Exception CommonException
 *
 * */

public synchronized static void start(Context context) throws CommonException
```

1.3.7 Get the Model Version

```
/**
 * @Function Get model version
 *
 * @Return Model version
 *
 * @Exception CommonException
 *
 * */

public synchronized static String getVersion() throws CommonException
```

1.3.8 Printer Reset

```
/**
 * @Function Printer reset
 *
 * @Tips The printer resets, all settings return to their default settings, and
the print buffer is cleared
 *
 * @Exception CommonException
 *
 * */

public synchronized static void reset() throws CommonException
```

1.3.9 Look for Black Labels

```
/**
 * @Function Look for black labels
 *
 * @Parameter
 * 1.search_disdance: 0-255
 * 2.walk_disdance: 0-255
 *
 * @Exception CommonException
 *
 * */

public synchronized void searchMark(int search_disdance, int walk_disdance) throws
CommonException
```

1.3.10 Paper Feed

```

/**
 * @Function Paper feed
 *
 * @Parameter
 * 1.line: line number [int > 0]
 *
 * @Exception CommonException
 *
 * */

public synchronized void walkPaper(int line) throws CommonException

```

1.3.11 Set the Print Alignment

```

/**
 * @Function Set the print alignment
 *
 * @Parameter
 * 1.mode:
 *     1.ALGIN_LEFT
 *     2.ALGIN_MIDDLE
 *     3.ALGIN_RIGHT
 *
 * @Exception CommonException
 *
 * */

public synchronized void setAlgin(int mode) throws CommonException

```

1.3.12 Set the Font Size

```

/**
 * @Function Set the font size
 *
 * @Parameter
 * 1.type:
 *     - Font size setting, 1-2 defaults to the No.2 font size
 *     - For single-byte characters, the font size of No.1 font size is 8*16
and the font size of No.2 font size is 12*24
 *     - For double-byte characters, the font size of No.1 font size is 16*16
and the font size of No.2 font size is 24*24
 *
 * @Exception CommonException
 *
 * */

public synchronized static void setFontSize(int type) throws CommonException

```


1.3.13 The Print Font Width and Height are Enlarged

```
/**
 * @Function The print font width and height are enlarged
 *
 * @Parameter
 * 1.widthMultiple: Default, 1
 * 2.heightMultiple: Default, 1
 *
 * @Exception CommonException
 *
 * */

public synchronized static void enlargeFontSize(int widthMultiple, int
heightMultiple) throws CommonException
```

1.3.14 Set the Print Grayscale Value

```
/**
 * @Function Set the print grayscale value
 *
 * @Parameter
 * 1.level: Need to check if your hardware supports it
 *
 * @Exception CommonException
 *
 * */

public synchronized void setGray(int level) throws CommonException
```

1.3.15 Add Print Content

```
/**
 * @Function Add print content
 *
 * @Parameter
 * 1.content: String content
 *
 * @Exception CommonException
 *
 * */

public synchronized void addString(String content) throws CommonException
```

1.3.16 Start Printing

```

/**
 * @Function Start printing, and the paper will not be taken after printing
 *
 * @Parameter
 * 1.entype: Such as ENGLISH, PERSIAN, KOREAN, JAPANESE, CHINESE
 *
 * @Exception CommonException
 *
 * */

public synchronized static void printString(String entype) throws CommonException

```

1.3.17 Paper Cut

```

/**
 * @Function Paper cut
 *
 * @Exception CommonException
 *
 * */

public synchronized static void paperCut() throws CommonException

```

1.3.18 Send Instructions to the Printer Device

```

/**
 * @Function Send instructions to the printer device
 *
 * @Parameter
 * 1.cmdStr: byte[] Command
 * 2.len: length
 *
 * @Exception CommonException
 *
 * */

public synchronized static void sendCommand(byte[] cmdStr, int len) throws
CommonException

```

1.3.19 Print the Picture

```

/**
 * @Function Print the picture
 *
 * @Parameter

```

```

    * 1.image: Image resource Bitmap object, width must be less than 384 pixels and
    * an integer multiple of 8, high pixels must be an integer multiple of 8
    *
    * */

    public synchronized void printLogo(Bitmap image, boolean isBuffer) throws
    CommonException

    //Barcode example
    Bitmap bitmap = CreateCode(barcodeStr, BarcodeFormat.CODE_128, 320, 176);
    if (bitmap != null) {
        mUsbThermalPrinter.printLogo(bitmap, true);
    }
    mUsbThermalPrinter.addString(barcodeStr);
    mUsbThermalPrinter.printString();

```

1.3.20 Release the Printer

```

/**
 * @Function Release the printer, which must be released when the printer is not
 * in use
 *
 * */

    public synchronized static void stop(Context context)

```

1.3.21 Set the Anti-white

```

/**
 * @Function Set the anti-white
 *
 * @Parameter
 * 1.isInverse: True or false
 *
 * @Exception CommonException
 *
 * */

    public static void setInverse(boolean isInverse) throw CommonException

```

1.3.22 Set the paper width

```

/**
 * @Function Set the paper width
 *
 * @Parameter

```

```
* 1.paperWidth: Paper width
*
* */

public static void setPaperWidth(int paperWidth)
```

1.3.23 Get the print count

```
/**
 * @Function Get the print count
 *
 * @Return Printed count
 *
 * */

public static long getPrintCount()
```

1.3.24 Get printer status

```
/**
 * @Function Get printer status
 *
 * @Exception CommonException
 *
 * */

public synchronized static int checkStatus() throws CommonException
```